FID: Fréchet Inception Distance

Measuring the fidelity and diversity of generated images, with industry examples and a hands-on activity

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Which is real & which one is AI-Generated?





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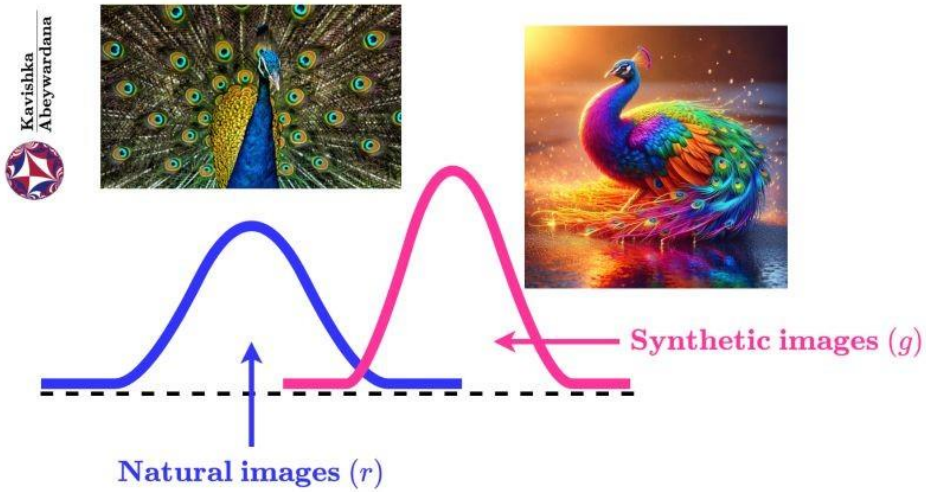
Which is real & which one is AI-Generated?





The Fréchet inception distance (FID) is a metric used to assess the quality of images created by a generative model, like a generative adversarial network (GAN)] or a diffusion model

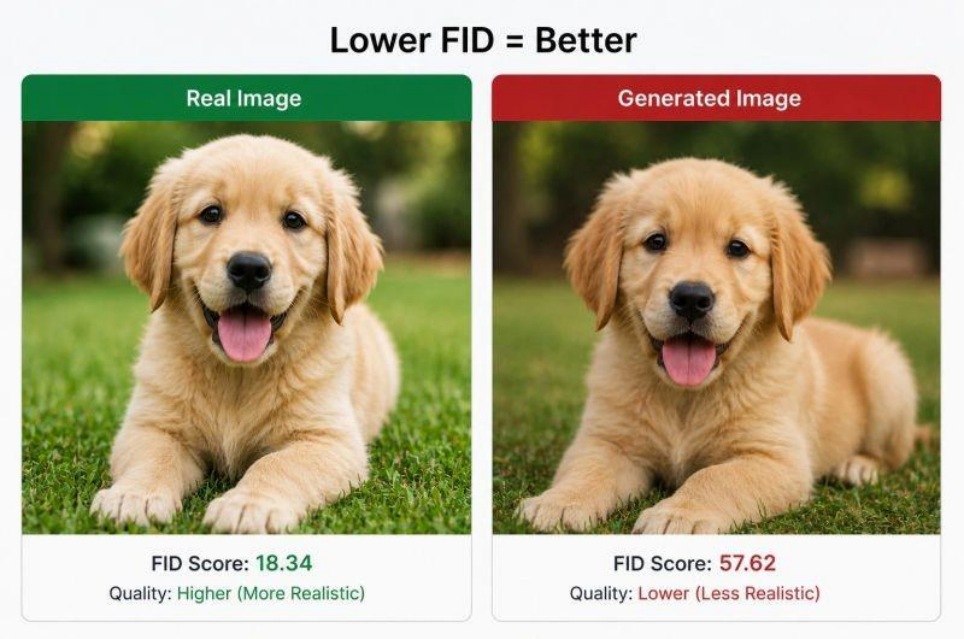
Real vs synthetic feature clouds



$$\mathrm{FID}(\mathcal{X}_r, \mathcal{X}_g) = \underbrace{\|\mu_r - \mu_g\|_2^2}_{\text{Average Difference}} + \underbrace{\mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right)}_{\text{Variabiity}}$$

Blue = natural images, pink = synthetic images. Left term: average difference. Right term: variability.

A good and a poor generated image with FID score



Careful: FID is calculated for a whole set of images (thousands), not for one picture. The per-picture scores here are example numbers only.

Let's understand What's FID?

1. What is FID (in simple words)?

- FID measures how close the distribution of generated samples is to the real data distribution in a feature space.
- It compares:
  - Mean (captures overall location / fidelity)
  - Covariance (captures spread, shape, and diversity)
- Lower FID = generated data is more similar to real data.

2. FID formula

$$\mathrm{FID} = \|\mu_r - \mu_g\|^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r \Sigma_g\right)^{1/2}\right)$$

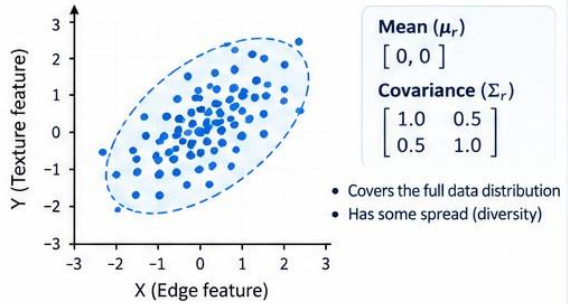
μ_r : mean vector of real data
μ_g : mean vector of generated data
Σ_r : covariance matrix of real data
Σ_g : covariance matrix of generated data
Tr : trace of a matrix (sum of diagonal elements)

3. X–Y vector image example

Let's imagine each image is represented by 2 features (X, Y) in a 2D space.
Each point = one image.

- X can represent "edge information" (e.g., how sharp the image is)
- Y can represent "texture information" (e.g., how detailed the image is)

A. Real Images (Ground Truth)

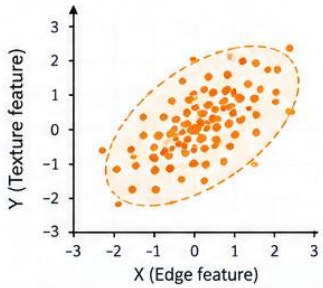


Mean (μ_r)
$[0, 0]$

Covariance (Σ_r)
$\begin{bmatrix} 1.0 & 0.5 \\ 0.5 & 1.0 \end{bmatrix}$

- Covers the full data distribution
- Has some spread (diversity)

B. Good Generated Images (Low FID)



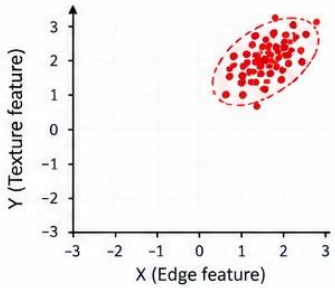
Mean (μ_g)
$[0.1, 0.1]$

Covariance (Σ_g)
$\begin{bmatrix} 0.9 & 0.4 \\ 0.4 & 0.9 \end{bmatrix}$

- Mean is close to real
- Covariance is similar
- Good fidelity and diversity)

FID ≈ 0.2 (Good)

C. Poor Generated Images (High FID)



Mean (μ_g)
$[2.0, 2.0]$

Covariance (Σ_g)
$\begin{bmatrix} 0.3 & 0.1 \\ 0.1 & 0.3 \end{bmatrix}$

- Mean is far from real
- Smaller spread (low diversity)
- Poor fidelity

FID ≈ 8.5 (Poor)

FID is an evaluation metric. Each image becomes one point with two features, X (edge) and Y (texture). Good generated images (B) sit close to the real cloud (A); poor ones (C) sit far away. Numbers are illustrative.

The formula

$$FID = ||\mu_1 - \mu_2||^2 + Tr(\Sigma_1 + \Sigma_2 - 2(\Sigma_1\Sigma_2)^{\wedge}½)$$

Ingredient	Describes	Real-life meaning
Mean	Where the cloud sits	Right kind of content and style
Covariance	How wide it is and which way it leans	How much variety there is

Centre term: $||\mu_1 - \mu_2||^2$ - Mean
Is the generated set in the right place? Grows with a colour cast or wrong content.

Spread term: Tr(…) - Covariance
Does it have the same width and shape? Grows with lost variety or extra noise.

Mean (μ): the centre of the cloud
What it is: the average of each feature over all images.
How to calculate it: add the values and divide by the number of images, n.
Example: 3 images with (X, Y) = (1, 2), (3, 3), (5, 7).
- X: (1 + 3 + 5) ÷ 3 = 3
- Y: (2 + 3 + 7) ÷ 3 = 4
- Mean **μ = (3, 4)**

Covariance (Σ): the spread and the tilt of the cloud
Covariance is a small table that answers two questions.

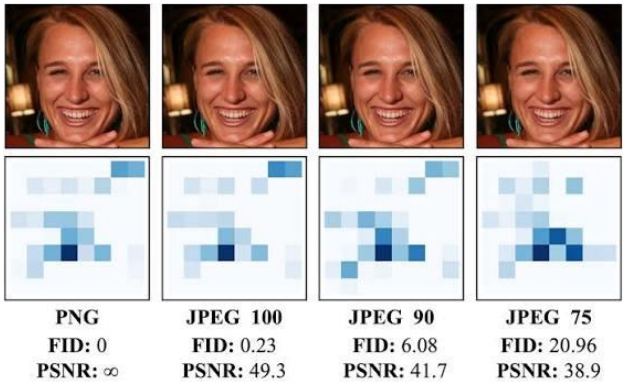
Question 1: how much does each feature vary on its own? (variance)
Gap of each image from the mean: X gaps are −2, 0, 2.
Square the gaps and add them: 4 + 0 + 4 = 8.
Divide by n − 1: 8 ÷ 2 = **4**.
Do the same for Y: gaps −2, −1, 3, squares 4, 1, 9, sum 14, ÷ 2 = **7**.

Question 2: do two features move together? (covariance)
Multiply the X gap by the Y gap for each image: (−2)(−2) = 4, (0)(−1) = 0, (2)(3) = 6.
Add them: 10. Divide by n − 1: 10 ÷ 2 = **5**.

μr:5 ,μg: 7 | Σr : 4 ,Σg: 9
Mean variance = $(5-7)^2 = (-2)^2 = 4$
Covariance = $4 + 9 - 2\sqrt{(4 \times 9)} = 13 - 2\sqrt{6} = 13 - 12 = 1$
FID = 4 + 1 = 5
Interpretation – the distributions differ in both their means & their covariances

Why do we need to score generated images?

- Generative models **(GANs**, diffusion) now make **faces, scans, satellite views** and **driving scenes**
- Looking at a few samples by eye is slow, subjective and misses missing variety
- Teams need one repeatable number to compare model A with model B
- **FID** is the most widely reported such number in image-generation research



The question
Does the generated set look like the real set, and vary like it?

The catch
One number is a screening tool, not proof of usefulness.

Two things to judge: fidelity and diversity

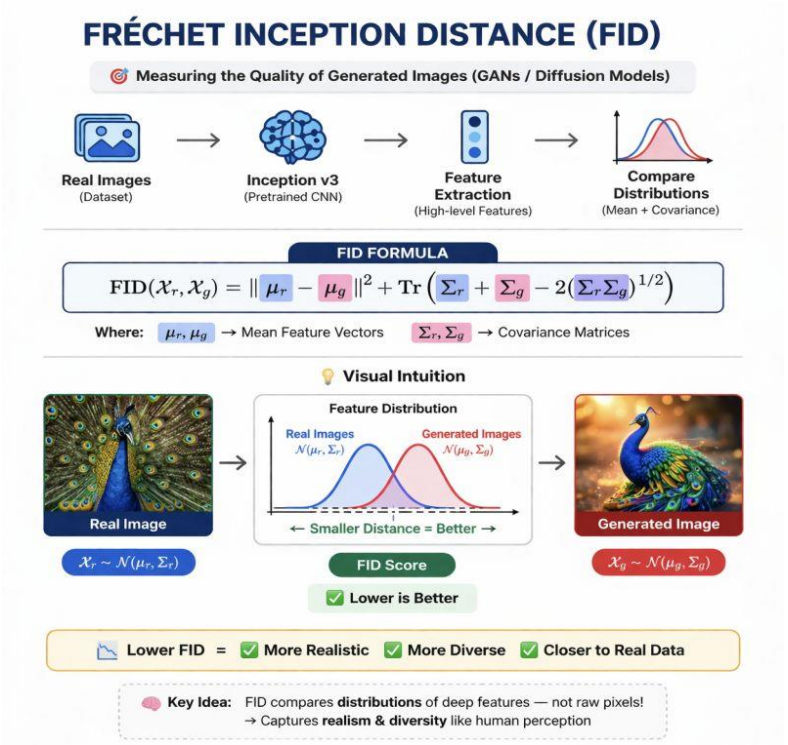
Fidelity (quality)

Does each image look real? Blur, noise and odd textures hurt fidelity.

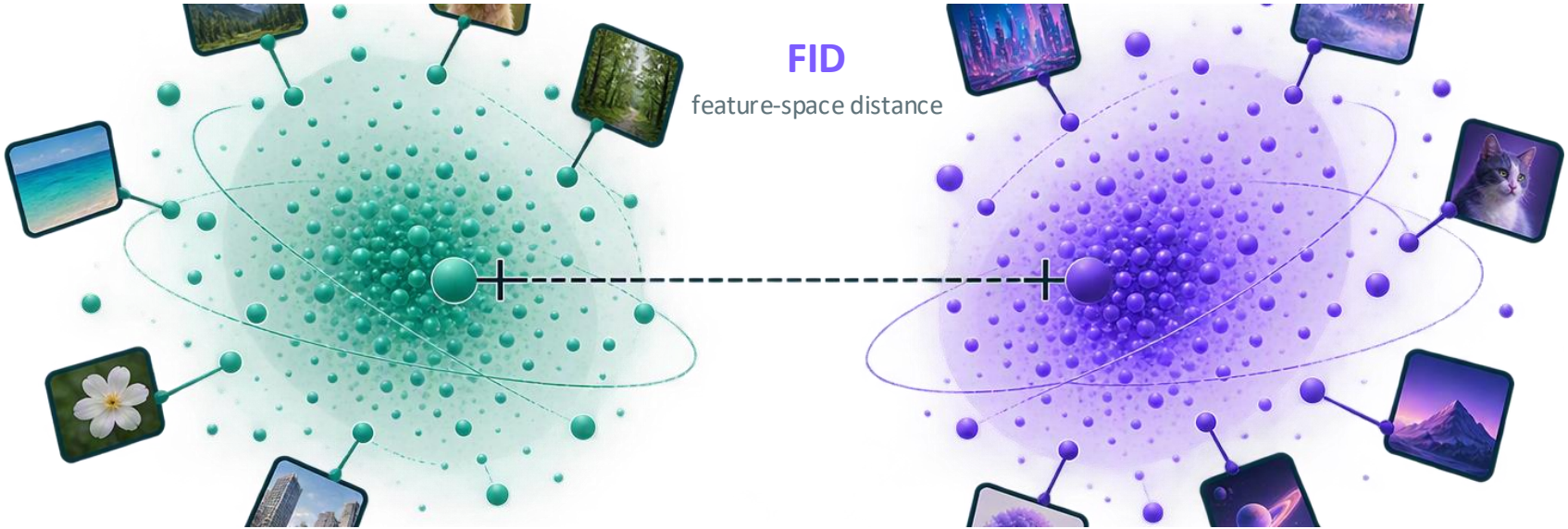
Diversity (variety)

Does the set cover the full range of real data? Repeating a few images is called mode collapse.

FID reacts to both, and blends them into a single number.



FID measures the gap between two image clouds



LOWER IS BETTER

FID = 0 indicates matching estimated feature distributions; it does not guarantee identical images.

COMPARE SETS

not individual pictures

USE ENOUGH IMAGES

10,000–50,000 per set

Where FID came from

Year	Metric	Idea	Gap
2016	Inception Score (IS)	Clarity and variety of generated images	Never looks at real images
2017	FID (Heusel et al.)	Distance between real and generated feature clouds	Assumes bell-shaped clouds
2019 to 2020	Precision/recall, density/coverage	Split quality from variety	Needs more care to read
2020 to 2024	KID, DINOv2-based FD, CMMD	Unbiased or modern-feature versions	Less common in older papers

FID in five steps

Compare like with like: same data, count, resize and extractor.

every image becomes a dot in a big space. A whole set of images becomes a **cloud of dots**. FID measures how far the real cloud is from the generated cloud.



01

Collect matched sets

Real + generated
≥10,000 each

FID compares sets

02

Inception-v3 feature extractor

Inception-v3
→ 2,048 features

Extract feature vectors from real images
Features capture content

03

Model both clouds

Compute μ + Σ for each set

Centre + diversity

04

Calculate the distance

Mean gap + covariance gap

Lower is better

05

Read the two terms

Compare feature-distribution means and covariance matrices.

Two matched sets → **2,048-D feature clouds** → one score

FID is model-agnostic in its application

FID is not domain-agnostic in its interpretation

FID is not a complete evaluation: a low score does not guarantee diversity, rare-class coverage, clinical validity, or fairness.

0 means a complete match, Lower generally indicates closer feature distributions.

$$FID = \|\mu_r - \mu_g\|^2 + Tr(\Sigma_r + \Sigma_g - 2 (\Sigma_r \Sigma_g)^{(1/2)})$$

- **r** = real images, **g** = generated images
- **μ** = mean (centre) of each set
- **Σ** = covariance (spread) of each set
- **Tr** = trace, the sum of the diagonal numbers
- **^(1/2)** = matrix square root (an exponent, not "times one half")

Inception-v3 turns each image into 2,048 meaningful numbers

One image in. One compact feature vector out.

01 PREPARE

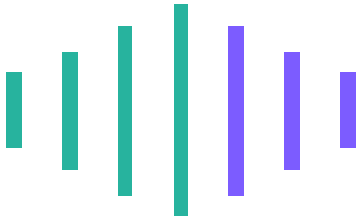
299 × 299

Resize every image consistently

The resize method affects the result.

→

02 EXTRACT



Inception-v3

trained on ImageNet
analyzes learned patterns

→

03 KEEP THE FINAL POOLING LAYER

2,048

numbers = one feature vector

[0.12 −0.84 0.37 … 0.31]

Different image → different vector

WHAT IT MEANS

The vector captures shapes, textures and objects — not raw pixel colours.

Similar-looking images produce similar vectors.

For fair FID comparisons, keep the resize method and extractor fixed. ImageNet features can misread specialized domains such as X-rays.

Reading the number, and the floor

FID on a well-known benchmark	Rough feel
Low single digits	Excellent
5 to 20	Good
20 to 50	Visible flaws or missing variety
Over 100	Poor

The floor
Score real images against other real images. It never reaches 0. It is the best value you can expect.

Only compare like with like
Same dataset, extractor, image count and resize method. FID 10 on faces is not FID 10 on X-rays.

What changes the score, not the images

Factor	Why it matters	What to do
Number of images	Few images push FID up	Same count for all sets; 50,000 for reports
Resize method	The **resize method changes features** (Parmar 2022)*	Use clean-fid, name the mode
Inception weights	TensorFlow and PyTorch versions differ	State which one
File format	JPEG compression shifts features	Same format for both sets
Reference set	Train, validation, test give different values	Say which one
Random seed	Different samples, different score	Repeat, report mean and spread

- ***Parmar(2022) -**Parmar, Zhang and Zhu (CVPR 2022) wrote *On Aliased Resizing and Surprising Subtleties in GAN Evaluation*. They found that popular libraries (PIL, OpenCV, PyTorch, TensorFlow) resize images in different ways, and that this changes FID scores noticeably. This paper is the reason the **clean-fid** library exists.

- before FID starts, every image is resized to 299 × 299 pixels, because Inception needs that size. Different software resizes in slightly different ways. The resized pictures are not identical, so Inception gives different feature numbers, and the FID changes even though the original images did not.

Sample size and the resize trap

Sample size

The covariance has 2,048 x 2,048 entries. Under about 2,048 images it cannot be fully estimated, so FID is noisy and biased upward. KID is unbiased and better for small sets.

Resize trap

The same images give different FID under different resize methods. Use one standard library (clean-fid) and report its mode.

Never compare a 500-image score with a 50,000-image score.

Limitations

The score inherits assumptions from both its **feature extractor and its single-number summary.**

REPRESENTATION LIMITS

01 **Gaussian assumption**

Several distinct clusters are flattened into one bell-shaped blob.

02 **ImageNet bias**

A few familiar classes can dominate the features and swing the score.

03 **Legacy extractor**

Inception-v3 tracks human ratings on modern diffusion outputs poorly; DINOv2 aligns better with human ratings.

EVALUATION BLIND SPOTS

04 **No semantic check**

FID cannot judge meaning, prompt following, or physical plausibility.

05 **Blind to copying**

A model that memorizes training images can still achieve an excellent score.

06 **One blended number**

Quality and variety are mixed, so the score cannot reveal which one failed.

BOTTOM LINE **Use FID as one signal—not a verdict. Add precision/recall, copying checks and human review.**

Medical images: which feature extractor?

Woodland et al., MICCAI 2024
Sixteen StyleGAN2 models on four medical imaging types, scored with several feature extractors and compared with expert ratings.

- ImageNet-based extractors were more consistent and closer to human judgement
- RadImageNet-trained extractors gave volatile rankings
- Lesson: test the extractor against experts in your own domain before trusting it

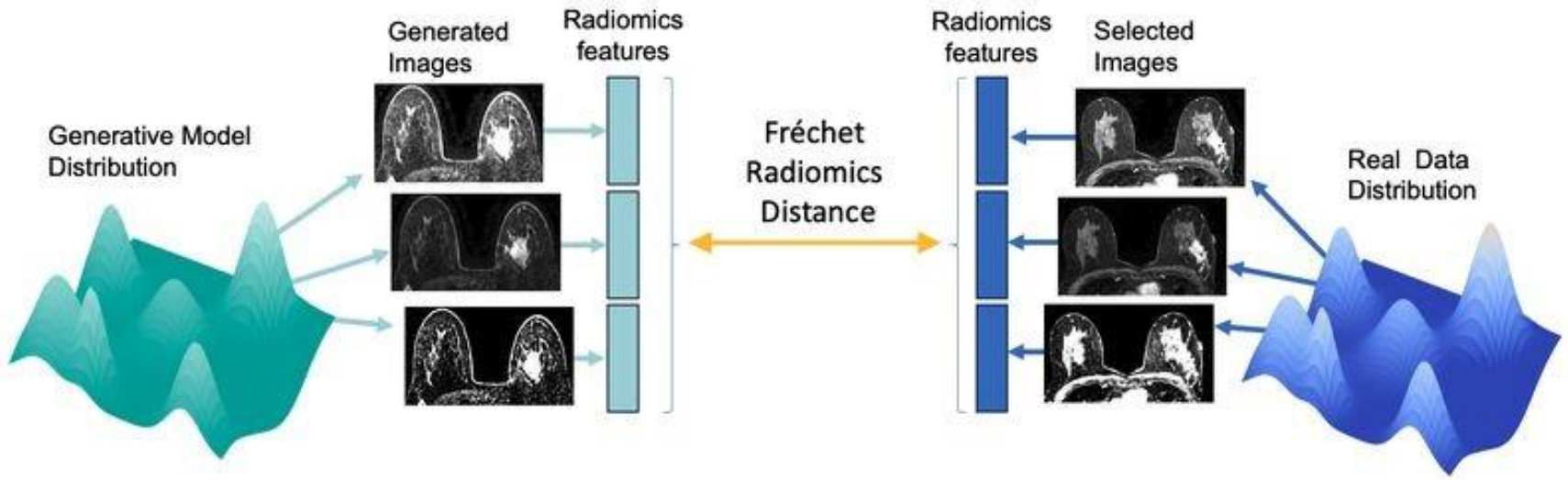
Source: arxiv.org/abs/2311.13717

Choose the metric based on what can go wrong

IF YOU NEED TO…	USE	BECAUSE
Compare small sets	**KID**	Unbiased when sample counts are limited.
Separate quality from variety	**Precision + recall**	Shows whether failures come from fidelity or coverage.
Reduce sensitivity to outliers	**Density + coverage**	A more robust diagnostic of the same two dimensions.
Assess modern or text-guided models	**DINOv2 FD / CMMD**	Modern features; CMMD also avoids the Gaussian assumption.
Evaluate another modality	**FVD / FAD / FCD**	Purpose-built for video, audio, and molecules.

BEST PRACTICE **Pair one distribution score with a diagnostic metric—and use human ratings for final judgement.**

Medical: Fréchet radiomics distance



The same Fréchet idea on breast MRI, but with radiomics features instead of Inception features. Only the feature extractor changes (picture supplied by the presenter).

Industry map: where FID is used

Industry	How common	Main use
AI image and video generation	Standard score	Compare one model with another
Medical imaging	Common in research	Check synthetic scans before training
Autonomous driving	Common in research	Check synthetic scenes and video
Disaster and conflict mapping	Used in papers	Check generated post-event images
Flood and climate visualization	Studied, opinions mixed	Judge realism of flood images
Deepfake and online safety	Idea stage	Describe how real a fake-maker is

Medical: extra scans for rare diseases

The situation

A hospital has only 200 scans of a rare eye disease. A computer program makes 2,000 more pictures to help train a diagnosis tool.

How FID helps

Before using the made-up scans, the team compares them with the real ones. A low score means they look and vary like the real ones.

A real example

Woodland et al. (2024) tested 16 picture-making programs on four kinds of medical images and compared FID with what experts thought.

Watch out

A low score does not prove a scan is medically correct. A program that simply copies real scans also scores well.



Source: arxiv.org/abs/2311.13717

Disaster response and conflict zones

The situation

After an earthquake, or in a conflict zone, there are few 'after' satellite pictures. A program makes 'after' pictures from 'before' ones, to train damage-spotting tools.

How FID helps

FID scores the made-up 'after' pictures against real ones.

A real example

A 2026 paper made post-event pictures for disasters and conflicts and reported scores 28.92 and 37.82. A drone-picture paper reported 23.7.

Watch out

A 2026 study found some programs make pictures look real but 'repair' damaged buildings. Also, scores from different papers cannot be compared.

Sources: ISPRS Archives XLIX-B3-2026; arxiv.org/abs/2603.20697

Flood and climate pictures

The situation

A city wants to show people how their street would look in a flood.

How FID helps

Researchers tried FID to check how real the made-up flood pictures look.

A real example

In 2019, the best FID version matched human opinions well, but not well enough alone. In 2021, ClimateGAN asked people instead, because FID needs many pictures.

Watch out

FID does not check whether water sits where water could really flow. Add a physical check.



Sources: arxiv.org/abs/1910.10143; arxiv.org/abs/2110.02871

Self-driving cars

The situation

Real driving data has few cases of night rain or fog. Teams make such scenes to train car-vision programs.

How FID helps

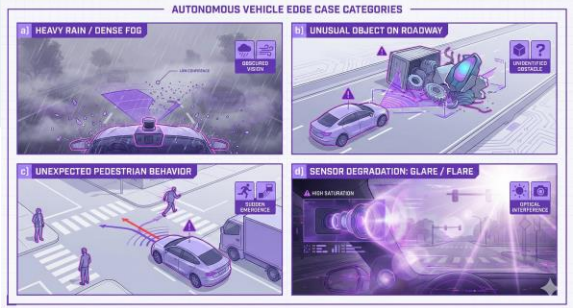
FID scores the pictures. FVD, its video cousin, scores the videos.

A real example

DrivingDiffusion (2023) reported FID 15.83 and FVD 332 on a public driving data set called nuScenes.

Watch out

A good overall score does not guarantee that cars and people look right. Newer papers also test a detection program on the made-up videos.



Source: arxiv.org/abs/2310.07771

Deep Fake Video Propaganda





Image credit - https://www.bbc.com/news/technology

Deepfakes and online safety

The situation

In 2022 a fake video of Ukraine's president spread online. Tools that spot fakes must keep up with new fake-makers.

How FID helps

FID shows how close a fake-maker is to real pictures.

A real example

An idea to test: if a new fake-maker is far from the one a detector learned from, the detector may fail more. The app shows this on dummy pictures.

Watch out

This is not proven. Measure real detector accuracy to know.

Hands-on activity: what we will build

make_data.py
Draws dummy images: coloured shapes, blurred, noisy and compressed copies, medical scans, war tiles and flood tiles.

run_fid.py
Sends each folder through Inception once, then calculates every FID score and saves them as CSV files.

app.py
The **Streamlit app**. It reads the CSV files and pictures and shows charts, summaries and the industry pages.

- **clean-fid** is a Python library (`pip install clean-fid`) made after the Parmar et al. 2022 paper. It calculates **FID and KID with one standard**, c**areful resize method,** so results are repeatable. It has resize modes (`clean`, `legacy_pytorch`, `legacy_tensorflow`) and a function `compute_fid(folder1, folder2)`.
- In this project it supplies the Inception feature extractor and the standard resizing. The distance itself is calculated in `run_fid.py`, which stays stable even with small image sets.
- Everything uses dummy pictures that imitate each field

Hands-on steps (VS Code or Colab)

VS Code on your laptop

1. File > Open Folder, pick fid_demo
2. Terminal > New Terminal
3. pip install clean-fid pillow numpy pandas streamlit torch
4. python make_data.py
5. python run_fid.py
6. streamlit run app.py

Google Colab (faster)

1. Runtime > Change runtime type > T4 GPU
2. Upload FID_Colab.ipynb and the 3 files
3. Run the cells in order
4. Download fid_results.zip
5. Unzip into fid_demo
6. streamlit run app.py

Hands-on tasks: predict, run, explain

Task	Predict	Look for
1 Floor: real A vs real B	Not 0	Your best possible value
2 Damage ladder: blur, noise, JPEG	FID rises each step	Which damage hurts most
3 Lost variety: 5 shapes to 1	FID rises	Spread term grows
4 Resize trap: clean vs legacy	Scores differ	Gap in points
5 Medical: memorized copy	Best score	Why FID cannot see copying
6 Flood: move the alert slider	Alert in a later week	First alert week

Checklist: using FID correctly

1. Fix and report extractor, resize method, format and image count
2. Compute the floor (real vs real)
3. Use 10,000+ images, ideally 50,000
4. Repeat with several seeds, report mean and spread
5. Add precision and recall
6. Check for copying (nearest neighbour)
7. Test the extractor against experts
8. Run a task test: train on synthetic, test on real

Common mistakes

- Reading lower FID as always better images
- Comparing across datasets or image counts
- Quoting a 500-image score like a 50,000-image score
- Trusting FID alone in medicine, safety or defence
- Forgetting the floor

Key takeaways

1
FID is the distance between the centre and spread of two image sets in feature space.

2
Its value depends on the extractor, image count and settings. Compare like with like.

3
It is a screening score. Confirm with a task test and expert review.

Small Quiz

1. What does a lower FID mean?

A) The generated images are further from real ones
B) The generated set is closer to the real set
C) The images are larger
D) The generator trained faster

2. What does FID compare?

A) One real picture with one fake picture
B) Pixel colours of two pictures
C) Feature distributions of a real set and a generated set
D) The file sizes of two folders

Small Quiz

3. Which network turns each image into 2,048 numbers?

A) ResNet-18
B) Inception-v3
C) GPT-4
D) VGG-16

4. A generator draws the same few images again and again. Which part of FID grows most?

A) The centre (mean) term
B) The spread (covariance) term
C) Neither
D) The file-size term

Small Quiz

5.What do you get when you compute FID between two sets of real images?

A) Exactly 0 every time
B) A small score above 0, called the floor
C) A negative number
D) An error

Thank you